

Interfacing with the Arduino : Module 4 Assignment

Smart Home Plant Monitoring and Watering System Using Arduino with WiFi Shield

Overview: This system is designed to monitor the moisture level of indoor plants and automatically water them when necessary. It uses an Arduino board with a WiFi shield to send real-time sensor data to a cloud-based dashboard and notify the user via email or app when attention is needed. This solution is ideal for individuals who travel frequently or tend to forget regular plant maintenance.

Components:

- **Arduino UNO** – the main microcontroller.
 - **WiFi Shield (e.g., ESP8266 or Arduino WiFi Shield)** – to connect the system to the internet.
 - **Soil Moisture Sensor** – measures the current moisture level in the soil.
 - **Water Pump (5V submersible)** – irrigates the plant when required.
 - **Relay Module** – acts as a switch to control the water pump.
 - **Power Supply** – provides power to the Arduino and peripherals.
 - **Water Reservoir** – holds water to be pumped.
 - **Plastic Tubing** – delivers water to the plant pot.
 - **Cloud Platform (ThingSpeak/Blynk/IFTTT)** – stores and displays data, and manages alerts.
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System Description:

1. Sensing Soil Moisture:

- The soil moisture sensor is embedded into the plant's soil.
- It outputs an analog value corresponding to the soil's moisture level.

2. Decision Making:

- The Arduino reads this value and compares it to a predefined threshold.
- If the soil is too dry (below threshold), the Arduino activates the relay.

3. Watering the Plant:

- The relay turns on the 5V water pump.
- Water is delivered from the reservoir through tubing into the soil.
- After a set duration, or once moisture improves, the pump is turned off.

4. Data Logging and Alerts:

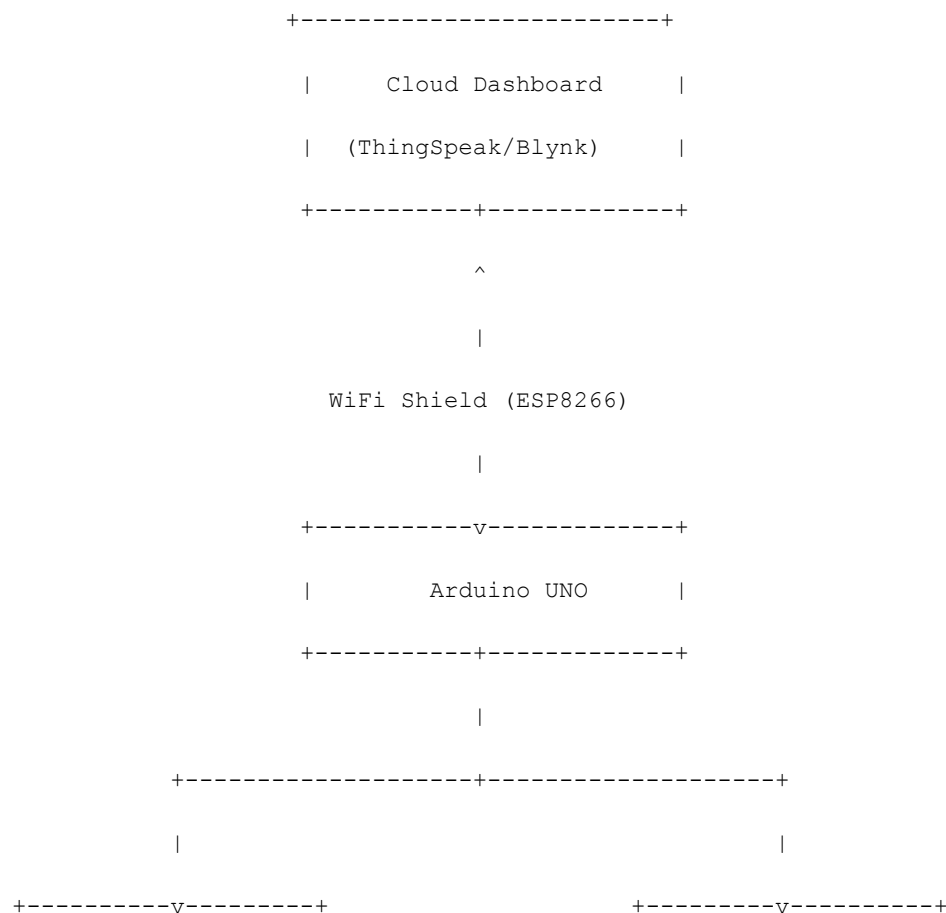
- Moisture data is sent to the cloud via WiFi.
- A user-friendly dashboard displays current status and history.
- If issues are detected (like water tank empty or faulty pump), an alert is sent to the user's phone/email.

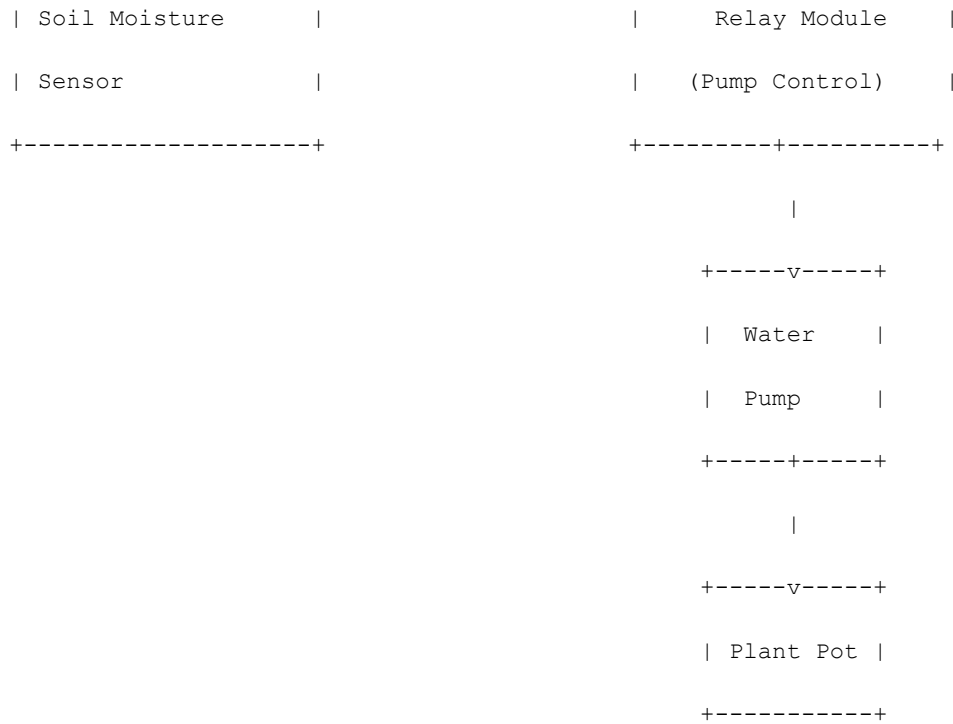
Need for WiFi Connection:

The WiFi connection plays a **critical role** in the functionality of this system:

- **Remote Monitoring:** View soil moisture levels from anywhere via smartphone or PC.
- **Alert System:** Receive push/email notifications if watering fails or soil stays dry too long.
- **Cloud Control:** Update moisture thresholds or watering durations remotely.
- **Smart Home Integration:** Connect with platforms like Alexa, Google Home, or IFTTT for automated actions or voice queries.

System Block Diagram:





EXAMPLE CODE

```

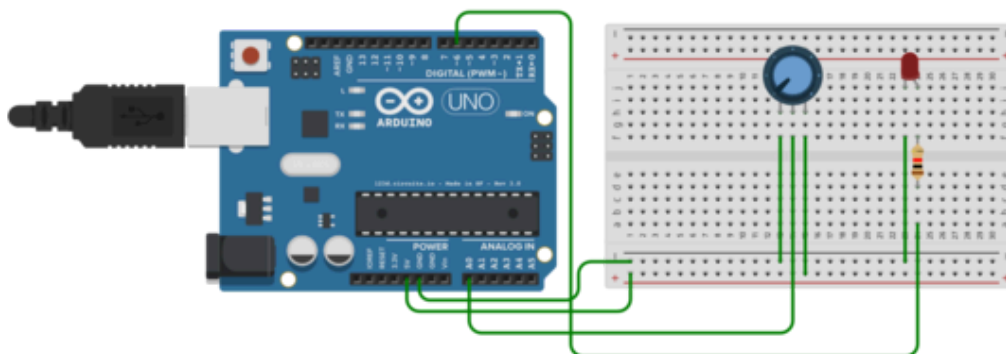
int sensorValue;           // value of potentiometer, 0 - 1023
int analogInPin = A0;      // Arduino pin connected to potentiometer
int analogOutPin = 6;      // Arduino pin connected to LED

void setup(){
  pinMode(analogInPin, INPUT);
  pinMode(analogOutPin, OUTPUT);
}

void loop(){
  sensorValue = analogRead(analogInPin);
  sensorValue = map(sensorValue, 0, 1023, 0, 255); // map to LED
  brightness range
  analogWrite(analogOutPin, sensorValue);          // set LED brightness
}
  
```

Additional Visual: Circuit for Brightness Control

Below is a basic Arduino circuit to control LED brightness using a potentiometer, which can also be used to adjust parameters like soil moisture thresholds in a physical interface.



Conclusion:

This smart plant monitoring and watering system highlights the power of Arduino combined with WiFi for home automation. It offers a low-cost, customizable solution for plant care, ensuring that users stay informed and their plants stay healthy – even when they're away from home.

The WiFi capability transforms a basic automation system into a smart, dynamic and adaptive system.